

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

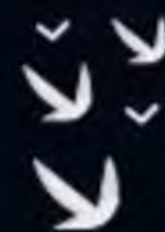
Pipeline Processing

Lecture No.- 2

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Recap of Previous Lecture



Doubt ?

Topic

Multiple Sector Access

Topic

Disk Cylinder

Topic

Disk Addressing



Topics to be Covered



Topic

Pipeline Processing

Topic

Pipeline Cycle Time

Topic

Speed Up



Topic : Parallel Processing

Parallel Processing: Simultaneous data processing

Types:

- Vector Processing
- Array Processing
- Pipeline Processing (Pipelining)



Topic : Pipeline Processing

- Pipelining is useful, When same processing is applied over multiple inputs





Topic : Pipeline Processing

- Technique to decompose a sequential process into sub-operations
- Sub-operations are performed in **segments** / *stages*
- Each segment perform it's respective suboperation in parallel
- **Task:** One operation performed in all segments



Topic : Example

Clock pulse number	Segment 1	Segment 2	Segment 3	Segment 4
1	Input 1			
2	I2	I1		
3		I2	I1	
4			I2	I1
5				I2
6				
7				
8				



Topic : General Consideration About Pipeline

- Consider a k segment pipeline with clock cycle time = t_p to perform n tasks

	no. of cycles	time
first task	k	$k * t_p$
remaining $(n-1)$	$n-1$	$(n-1) t_p$
Total	$(k+n-1)$	$(k+n-1) t_p$



Topic : General Consideration About Pipeline

- Consider a non-pipeline system that takes t_n time to perform a task

$$n \text{ Task time} = n * t_n$$



Topic : General Consideration About Pipeline

- Performance of a pipeline is given by **Speed up ratio**.

$$S = \frac{n * t_n}{(k + n - 1) t_p}$$

when $n \gg k$

$$S_{\text{ideal or max}} = \frac{t_n}{t_p}$$

special case when $t_n = k t_p$

$$S_{\text{ideal}} = k$$

Ideal Condition:-

Ignore first $(k-1)$ cycles to fill the pipe, hence pipeline will take 1 cycle per input and n cycles total for n inputs.

$$n \text{ tasks time} \Rightarrow \begin{array}{c|c} \text{cycles} & \text{time} \\ \hline n & n * t_p \end{array}$$

$$S_{\text{ideal}} = \frac{\cancel{n} * t_n}{\cancel{n} * t_p} = \frac{t_n}{t_p}$$

- #Q. A non-pipeline system takes 50 ns to process a task. The same task can be processed in a six-segment pipeline with a clock cycle of 10ns.
1. Determine the speedup ratio of the pipeline for 100 tasks.
 2. What is the maximum speedup that can be achieved?

$$t_n = 50 \text{ ns}$$

$$k = 6$$

$$t_p = 10 \text{ ns}$$

$$1. \quad S = \frac{100 * 50}{(6 + 100 - 1) 10} = 4.76$$

$$2. \quad S_{ideal} = \frac{50 \text{ ns}}{10 \text{ ns}} = 5$$

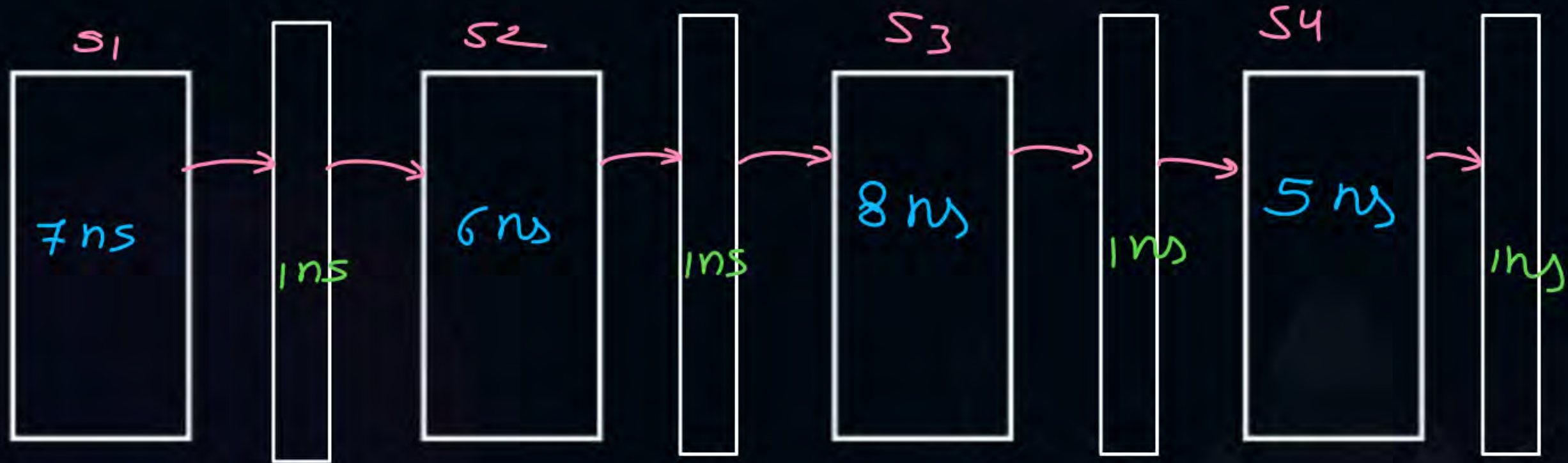
when in Questⁿ $\Rightarrow$ ideal speed up asked
or
ideal condition mentioned
or
value of n is not given
or
one input takes one cycle

$\Rightarrow$ ideal condition formula



Topic : Synchronous Pipeline

Intermediate buffer or reg. $\leftarrow$ used only in pipeline



$$t_p = \max \text{ of segment delays} = \max(7, 6, 8, 5) = 8 \text{ ns}$$

$$t_p = \max \text{ of seg. delays} + \text{Reg./buffer delay} = \max(7, 6, 8, 5) + 1 = 9 \text{ ns}$$



Topic : Cycle Time in Synchronous Pipeline

$$\begin{aligned}t_n &= \text{sum of seg. delays w/o Reg.} \\&= (7 + 6 + 8 + 5) \\&= 26 \text{ ns}\end{aligned}$$

#Q. Consider 6 segment pipeline with segment delay of segments as 20ns, 26ns, 21ns, 21ns, 24ns and 28ns respectively. Calculate processing time of pipeline for 1000 tasks?

$$k = 6$$

$$n = 1000$$

$$t_p = \max(20, 26, 21, 21, 24, 28) = 28 \text{ ns}$$

$$\begin{aligned} \text{pipeline time} &= (6 + 1000 - 1) 28 \\ &= 28140 \text{ ns} \end{aligned}$$

$$\begin{aligned} t_h &= 20 + 26 + 21 + 21 \\ &\quad + 24 + 28 = 140 \end{aligned}$$

$$\begin{aligned} S &= \frac{1000 \times 140}{(6 + 1000 - 1) 28} \\ &= 4.975 \end{aligned}$$

#Q. The time delay of the four segments in pipeline are as shown follows:

$t_1 = 50 \text{ ns}$, $t_2 = 30 \text{ ns}$, $t_3 = 95 \text{ ns}$, and $t_4 = 45 \text{ ns}$.

The interface registers delay time $t_r = 5 \text{ ns}$.

How long would it take to process 100 tasks in the pipeline?

$$t_p = \max(50, 30, 95, 45) + 5 = 100 \text{ ns}$$

$$n = 100$$

$$k = 4$$

$$\begin{aligned} \text{pipeline time} &= (4 + 100 - 1) 100 \\ &= 10300 \text{ ns} \end{aligned}$$

$$\begin{aligned} t_n &= 50 + 30 + 95 + 45 \\ &= 220 \text{ ns} \end{aligned}$$

$$\begin{aligned} S &= \frac{220 * 100}{(4 + 100 - 1) 100} \\ &= 2.1359 \end{aligned}$$

$\Rightarrow$ pipeline design must have almost equal delays for segments to provide better results.

#Q. How can we reduce the total time about the one-half of the time calculated in above question?

The seg. 3, if possible, divided into 2 segments $\begin{matrix} \nearrow 50 \text{ ns} \\ \searrow 45 \text{ ns} \end{matrix}$

new pipeline $\Rightarrow$ 5 segments

seg 1 $\Rightarrow$ 50 ns

seg 2 $\Rightarrow$ 30 ns

seg 3.1 $\Rightarrow$ 50 ns

seg 3.2 $\Rightarrow$ 45 ns

seg 4 $\Rightarrow$ 45 ns

$$t_p = \max(50, 30, 50, 45, 45) + 5 \\ = 55 \text{ ns}$$

$$\text{pipeline time} = (5 + 100 - 1) 55 \text{ ns} \\ = 5720 \text{ ns}$$

$$\text{Speed up of new pipeline} = \frac{220 * 100}{5720} = 3.84$$

speed up achieved by new pipeline as compared to old pipeline.

$$= \frac{10300}{5720}$$

$$= 1.8$$

Ques) 5 stage pipeline

$S_1 \Rightarrow 16 \text{ ns}$

$S_2 \Rightarrow 20 \text{ ns}$

$S_3 \Rightarrow 25 \text{ ns}$

$S_4 \Rightarrow 40 \text{ ns}$

$S_5 \Rightarrow 22 \text{ ns}$

Reg $\Rightarrow 2 \text{ ns}$

$n = 100$ tasks

$\Rightarrow$ new
seg S_4 is divided
into 2
 $\swarrow \searrow$
delay 25 ns delay 16 ns

Ques

Speed up of new
pipeline as compared
to old ?

old	new
$k = 5$	$k = 6$
$t_p = 40 + 2 = 42 \text{ ns}$	$t_p = 25 + 2 = 27 \text{ ns}$
$n = 100$	$n = 100$
$\text{time} = (5 + 100 - 1) \cdot 42$ $= 4368 \text{ ns}$	$= (6 + 100 - 1) \cdot 27 \text{ ns}$ $= 2835 \text{ ns}$

$$\text{speed up of new pipeline} = \frac{4368}{2835} = 1.54$$

cycle time of pipeline:-

$$\text{cycle time} = \frac{1}{\text{pipeline clock rate}}$$

#Q. Consider a non-pipelined processor with a clock rate of 8 gigahertz and average cycles per instruction of 5. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 5 gigahertz. Assume that there are no stalls in the pipeline. The speed up achieved in this pipelined processor is _____.

non-pipeline:-

$$\text{cycle time} = \frac{1}{8} \text{ ns}$$

$$t_n = 5 * \frac{1}{8} \text{ ns} = 0.625 \text{ ns}$$

pipeline:-

$$\text{cycle time } (t_p) = \frac{1}{5 \text{ GHz}} = 0.2 \text{ ns}$$

$$S = \frac{0.625 \text{ ns}}{0.2 \text{ ns}} = 3.125$$



Topic : Latency and Throughput

non-pipeline $\Rightarrow t_n$

$$\text{pipeline} = k * t_p$$

latency $\Rightarrow$ time required to perform one task.

In general

$$k * t_p \geq t_n$$

no. of operations performed per unit time

non pipeline:-

in t_n time no of tasks = 1

$$\text{in } 1 \text{ ——— } || \text{ ——— } = \frac{1}{t_n}$$

Throughput

pipeline:-

in $(k+n-1)t_p$ time no. of tasks = n

$$\text{in } 1 \text{ ——— } || \text{ ——— } = \frac{n}{(k+n-1)t_p}$$

throughput in ideal condition = $\frac{1}{t_p}$

#Q. Consider 6 segment pipeline with segment delay of segments as 20ns, 26ns, 21ns, 10ns, 18ns and 20ns respectively. The intermediate buffer delay is 5ns.

Consider that the system is used for performing 100 tasks.

1. What is the latency of non-pipeline system 115 ns
2. What is the latency of pipeline system $6 * 31 = 186 \text{ ns}$
3. What is the throughput of pipeline system $\frac{100}{(6+100-1)31} = 0.0307 \text{ per ns}$
4. What is the throughput of pipeline system in ideal case $\frac{1}{31} = 0.032 \text{ per ns}$

$$t_n = 20 + 26 + 21 + 10 + 18 + 20 = 115 \text{ ns}$$

$$t_p = 26 + 5 = 31 \text{ ns}$$

#Q. The stage delays in a 4-stage pipeline are 60ns, 50ns, 55ns and 80ns. The last stage (with delay 80ns) is replaced with a functionally equivalent design involving two stages with respective delays 60ns and 35ns. The throughput increase of the pipeline is 33.33 percent?

<u>old</u>	<u>new</u>
$t_p = 80\text{ns}$	$t_p = 60\text{ns}$
throughput = $\frac{1}{80}$	$= \frac{1}{60}$

n not given hence use ideal case

Throughput increase

$$= \frac{\frac{1}{60} - \frac{1}{80}}{\frac{1}{80}} * 100\%$$

$$= \left(\frac{4}{3} - 1 \right) * 100\% = 33.33\%$$

- #Q. Consider a 6-stage pipeline with delays 2, 4, 3, 5, 3 and 4 ns. This pipeline is upgraded to a new 8-segment pipeline in which each segment delay is 2 ns.
1. How much time is saved using new pipeline over old one? For 100 inputs?
 2. What is the speed up of new pipeline as compared to old pipeline?



Topic : Instruction Pipeline

If pipeline processing is implemented on Instruction Cycle

Instⁿ fetch
Instⁿ decode
E.A. calculatⁿ
operand fetch
Execution
write back result



Topic : Instruction Pipeline



Assume a 5 stage pipeline

- IF: Instruction Fetch
- ID: Instruction Decode & Address Calculation
- OF: Operand Fetch
- EX: Execution
- WB: Write Back



2 mins Summary



Topic

Pipeline Processing

Topic

Pipeline Cycle Time

Topic

Speed Up





Happy Learning

THANK - YOU

